GEOPHYSICAL RESEARCH LETTERS, VOL. 20, NO. 16, PAGES 1667-1670, AUGUST 20, 1993

PALEOMAGNETIC AND ROCK MAGNETIC INVESTIGATIONS OF THE CHANGXING PERMIAN-TRIASSIC SECTION, ZHEJIANG PROVINCE, CHINA

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Abstract. Paleomagnetic and rock magnetic experiments on the Triassic Yinkeng Formation and the Permian Changxing Formation sampled from the Permian-Triassic type section, Changxing County, reveal that a high temperature primary direction is isolated in dolomitic samples from the Early Triassic but the Late Permian limestones samples appear to be remagnetized during the Jurassic. A Permian-Triassic pole previously reported for the Changxing Formation from this section (Lin et al., 1985) is not in agreement with results from this study or other poles for that time interval reported for the Yangtze Block and may be remagnetized.

Introduction

The Changxing Permian-Triassic type section is located in the Baoqing Quarry, Changxing County, about 80 km north of Hangzhou (Figure 1). Exposed in the quarry is 50 meters of irregularly bedded greyish limestones with some interbedded grey mudstones and shales, and dolomites and limestones in the late part of the Yinkeng Formation. The boundary between the Late Permian Changxing Formation and the Early Triassic Yinkeng is marked by a greyish-yellowish boundary clay layer. The section was sampled by Lin and others (1985) and the results were used in the calculation of a Permian-Triassic pole for the Yangtze Block.

This section was also sampled by Li and Wang (1987) and they were able to resolve a magnetic stratigraphy though Lin (1984) reported only one polarity from the section. Recently it has been shown that Lin's pole from the Changxing section may be remagnetized (Steiner et al., 1989; Dobson and Mauritsch, 1989). In the most recent and comprehensive compilation of Chinese results from the Permian to Present, Enkin et al. (1992) use the results from 16 samples at the very top of this section (from the study of Li and Wang) in calculations of a mean Early Triassic paleopole for the Yangtze Block. These were the only results from outside of the south-central Sichuan Province area used by Enkin et al. (1992) in the calculation of the Early Triassic direction for the block and are, therefore, very important. For this reason, we present here new results from our study of the Changxing section and examine the rock magnetic properties of samples from the section.

Results

The Changxing Formation

Fifty-five samples were taken from the type section in Baoqing quarry - 32 from three sites with slightly differing bedding attitudes (45dip/330az, 34/318, 39/319) from the Late Permian Changxing Formation and 23 from three sites with differing bedding attitudes (33.5/319, 56.5/305, 46/44) from the Early Triassic Yinkeng Formation as part of a tectonic study. The remanent magnetizations for all samples were analyzed using standard paleomagnetic techniques (Zijderveld, 1967; Kirschvink, 1980; Fisher, 1953). Two components of magnetization are seen in the Zijderveld diagrams during thermal demagnetization of samples from the Changxing Formation (Figure 2a). The low temperature component is mainly along the present field direction and persists to about 250°C to 300°C. The high temperature component decays to the origin by 450°C. Of the 32 Permian samples, a stable, characteristic high-temperature direction was recognized in only 13 samples. Equal area projections of the high temperature ChRM directions for these 13 samples yield an *in situ* declination of 65.7° and inclination of 44.8° with an α_{95} of 5.2, which is significantly different from the present field/dipole field directions. The bedding corrected, normal polarity mean declination of 27.6° and a mean inclination of 40.9° with an α_{95} of 6.5° gives a paleopole of 64.5°N, 219.7°E which is not consistent with the Late Permian reference pole of Enkin et al. (1992) from the Yangtze Block (Figure 3a, Table 1). This is an indication that the formation is likely remagnetized.

Isothermal remanent acquisition (IRM) experiments were conducted on samples from the type section in order to determine which mineral phase is carrying the ChRM component. IRM acquisition curves for samples from the Changxing Formation reveal the presence of both low and

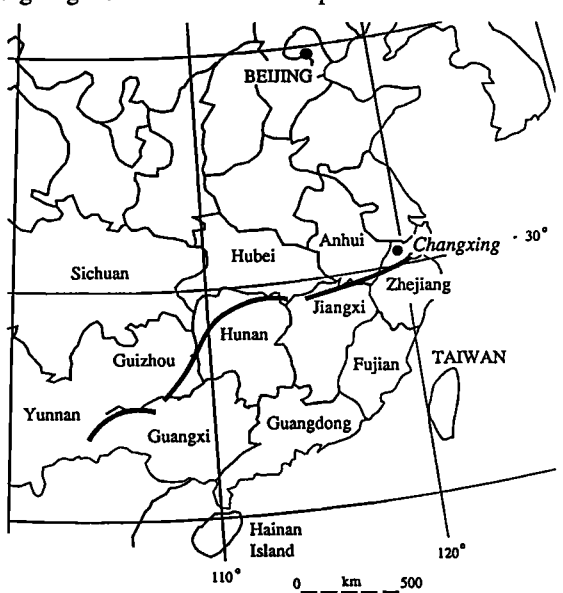


Fig. 1. Map of China showing the location of the Changxing section and the suture between the Huanan and Yangtze blocks according to Hsü et al., 1990.

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Paper number 93GL01242
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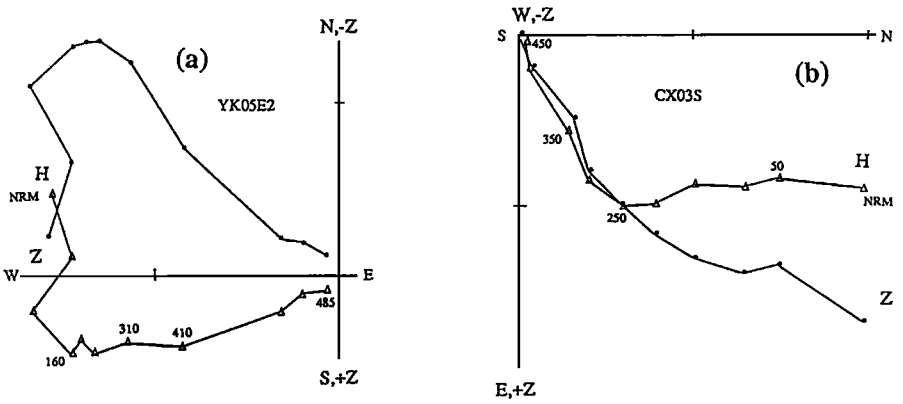


Fig. 2. Representative Zijderveld diagrams during thermal demagnetization for the (a) Yinkeng Formation and (b) Changxing Formation. Temperature in °C. Triangles are the horizontal projections and circles are the vertical projections.

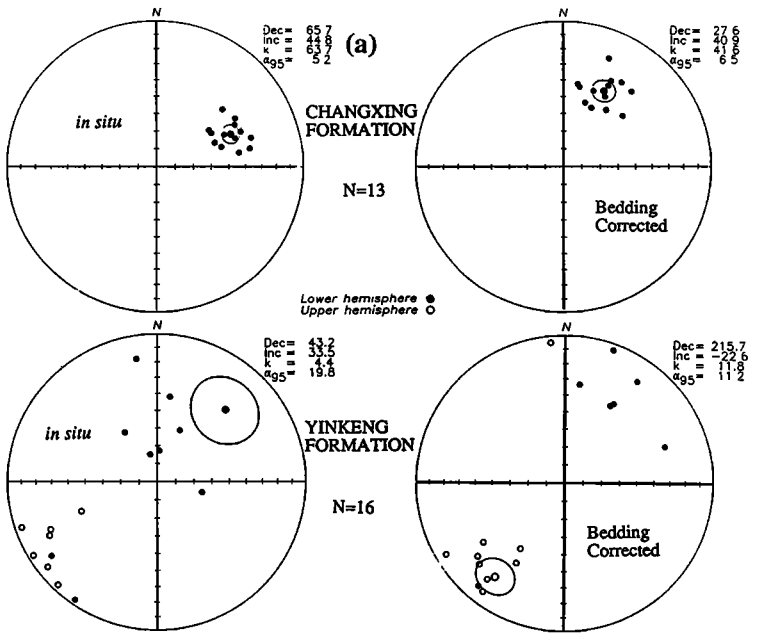


Fig. 3. (a) Equal area projections of stable directions for the Changxing Formation (b) Equal area projections of stable directions for the Yinkeng Formation.

TABLE 1. Pole positions derived from the formation means. N = number of samples

		Before Bedding Correction				After Bedding Correction			
FORMATION	N	LAT.	LONG.	A95	k	LAT.	LONG.	A95	k
Changxing Formation	13	33.0°N	193.8°E	5.2	63.7	64.5°N	219.7°E	6.5	41.6
Yinkeng formation	16	49.0°N	216.6°E	19.8	4.4	51.8°N	232.1°E	11.2	11.8

high coercivity minerals. Thermal demagnetization of the IRM indicates that both hematite and magnetite may be present though the blocking temperature spectra appear to completely overlap. There is no indication of the presence of goethite with low unblocking temperatures in the demagnetization curves (Figure 4a).

The Yinkeng Formation

Of the 23 samples from the Early Triassic Yinkeng formation, 16 had stable, high temperature ChRM components. The remaining samples, collected from limestone beds, had very weak NRMs which were completely overprinted by the present field. All 16 stable samples were recovered from dolomitic layers. The behaviour during demagnetization was similar to the Changxing Formation with a present field component persisting to about 250°C (Figure 2b). Equal area projections of the high temperature ChRM directions reveal a bedding corrected reversed mean declination of 215.7° and inclination of -22.6° with an α_{95} of 11.2, which yields a paleopole at 51.8°N, 232.1°E (Figure 3b). This formation passes the fold test at the 99% confidence level indicating that the remanence was acquired before the Early Cretaceous (the age of folding). The paleopole is in excellent agreement with the results of Li and Wang for this same section (1987; Pole = 51.1°N, 230.3°E). Those results were the used by Enkin et al. (1992) in their calculation of an Early Triassic pole for the Yangtze Block and the pole from our study falls directly on that pole path (Figure 5).

Further, analysis of the α_{95} confidence circles for the Yinkeng Formation and the Changxing Formation, indicates that the mean directions from the two formations are not

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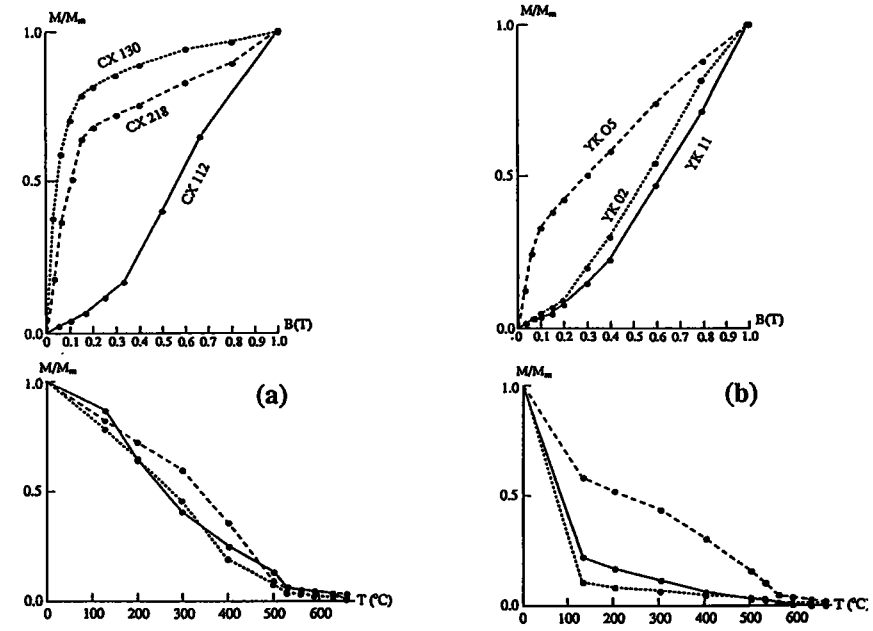


Fig. 4. (a) IRM acquisition and thermal demagnetization curves for samples from the Changxing Formation (b) IRM acquisition and demagnetization curves for samples from the Yinkeng Formation.

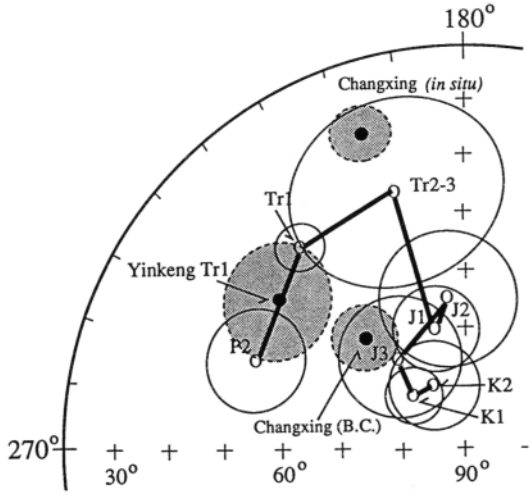


Fig. 5. Comparison of the paleopoles from this study with the APWP of Enkin et al. (1992). B.C. = bedding corrected pole position.

antipodal and probably preserve characteristic remanent magnetizations of differing ages.

IRM experiments were also performed on samples from the Yinkeng Formation and a dominance of high coercivity minerals is seen (Figure 4b). A large decrease in intensity at the 130°C thermal demagnetization step is consistent with the presence of goethite. Again, in some of the samples, a low coercivity component is also present. Further thermal cleaning shows that these samples lose most of the remainder of the IRM by 560°C which indicates the presence of magnetite. However, some intensity persists to temperatures of 630°C to 660°C and is likely indicative of the presence of minor amounts of hematite.

Discussion

The results from the Permian Changxing Formation and the Triassic Yinkeng Formation at the type section, analyzed separately, reveal that the Early Triassic results are in excellent agreement with the Early Triassic Paleopole position of Li and Wang (1987) but the Permian results appear to be remagnetized.

Results from the Late Permian Changxing Formation from the type-section at Baoqing quarry yield a pole position which is significantly displaced from the Late Permian reference pole for the Yangtze Block and fall closer to the Middle to Late Jurassic section of the APWP (Enkin et al., 1992). Both pre and post-folding Jurassic remagnetizations have been documented in south China (Dobson and Heller, 1992) and, although k decreases upon bedding correction, this remagnetization appears to be pre-folding since the *in situ* pole position (33.0°N, 193.8°E) falls off of the APWP. The large A95 of the Middle to Late Triassic pole position (Enkin et al., 1992), however, and the recent discrepancies of Triassic results for the Yangtze Block (Steiner et al., 1989; Dobson and Heller, 1993) mean that a post-folding, Middle to Late Triassic remagnetization should not be completely ruled out.

The question then arises - why are the Yinkeng sediments apparently unaffected by the remagnetization that affected the Changxing Formation? Results of the IRM studies reveal that the only significant difference in the magnetic minerals from the two formations is the presence of goethite in the Yinkeng Formation. There is evidence for the presence of magnetite as well as hematite in both formations, though the relative abundances and grain sizes may play a role in the stability of the signals. The overlapping of the blocking temperature spectra, however, make it difficult to determine the relative contributions of each (Figure 4).

All of the stable Yinkeng samples come from the dolomitic layers in the section, while the samples from the Changxing Formation and the unstable Yinkeng samples come from limestones. The only difference seen in the IRM experiments between the dolomites and limestones is reflected in the presence of goethite in the dolomites. Its origin is unclear. It seems then, that the dolomitization of these sediments may be responsible for the preservation of the Early Triassic magnetization since the formation of dolomite can, in many cases, greatly decrease the porosity of a rock unit. This reduction in porosity may have prevented fluids responsible for the Jurassic remagnetization events from passing through the dolomite, thus enabling the preservation of the Early Triassic signal in those sediments.

Conclusions

In light of this more detailed analysis of the type section, the likelihood arises that the pole reported by Lin and others

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(1985) from the Changxing section is remagnetized. This would explain the recently suggested discrepancies in their pole with other results for the same time period (Steiner et al., 1989; Dobson and Mauritsch, 1989; Li, 1988). Any calculation of a mean Permian - Triassic pole for the Yangtze Block therefore, should exclude that pole.

Recent results by Dobson and Heller (1993) have shown that rotations may have affected the central region of the Yangtze Block making it difficult to determine whether the Early Triassic pole from the APWP of Enkin et al. (1992) is correct. Their (Enkin et al.) Early Triassic pole is based almost exclusively on results from central Sichuan Province or its border - the only exception being the work done by Li and Wang (1987) on the Changxing section. For this reason, results from the Changxing Section in the eastern part of the block are central to the argument that the Enkin et al. APWP is correct for the Early Triassic.

The extremely close agreement of the Early Triassic pole of this study with the results of Li and Wang (1987) and the Yangtze Block APWP, along with the fact that the Yinkeng Formation passes the fold test, is an indication that, though the sample number is small, the Early Triassic Yinkeng Formation at the Changxing section provides a reliable result for the eastern Yangtze Block. The agreement of these results from the eastern part of the block with results from the central part indicate that these areas have not been affected by local thrust sheet rotations and are representative of the block as a whole.

Acknowledgments. We would like to thank Profs. K.J. Hsü, Sun Shu, C.McA. Powell, and D. Fang for their help with this project and Drs. Chen Haihong, Shui Tao, and Lin Jin-lu for their help with the field work. We also thank Randy Enkin and an anonymous reviewer for their comments, which led to an improvement of the final manuscript. This work is supported by Swiss National Funds Project no. 21-29834.90 and Australian Research Council grant no. A39030719.

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(Received September 9, 1992;
revised April 21, 1993;
accepted April 22, 1993)

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